PROJECT REPORT:

Student Performance Dashboard

Subject: Data Analytics

Subject Code: 25CAH-727

Submitted By

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Under the guidance of

Ms. Sweta

In the partial fulfilment for the award of the degree of

MASTER OF COMPUTER APPLICATIONS

Artificial Intelligence and
Machine Learning



Chandigarh University

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CERTIFICATE

This is to certify that the project report entitled “Student Performance Dashboard”, submitted by Aditya Aman (Roll No. 25MCI10017) in partial fulfilment of the requirements for the award of the degree of Master of Computer Applications (Artificial Intelligence and Machine Learning) at Chandigarh University, is a record of work carried out under my guidance and supervision.

To the best of my knowledge, the work presented in this report is original and has not been submitted elsewhere for the award of any other degree or diploma.

Date: _____ Signature: _____

Place: Chandigarh University

Ms. Sweta
Project Guide

DECLARATION

I, Aditya Aman (25MCI10017), a student of Master of Computer Applications (Artificial Intelligence and Machine Learning) at Chandigarh University, hereby declare that the project report entitled “Student Performance Dashboard” is an original work carried out by me under the guidance of Ms. Sweta.

The data used in this project is a sample dataset created for academic demonstration and does not represent real students. The content of this report has not been submitted, in part or in full, to any other university or institution for the award of any degree or diploma. All sources of information used have been duly acknowledged in the references.

Date: _____

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ACKNOWLEDGEMENT

I would like to express my sincere gratitude to Chandigarh University and the University Institute of Computing (UIC) for providing me with the opportunity, academic environment, and resources required to undertake this project, “Student Performance Dashboard: A Data Analytics Workbook for Academic Performance Monitoring.” The knowledge gained during my MCA programme has helped me connect theoretical concepts of data analytics with a practical, working solution.

I would like to extend my heartfelt appreciation to my project guide, Ms. Sweta, and the faculty members for their valuable guidance, encouragement, and constructive suggestions throughout this project. Their support helped me understand the different stages of an analytics project, including problem definition, data preparation, formula design, visualisation, validation, and interpretation of results.

I am also thankful to my classmates and friends for their suggestions and discussions during the development of the dashboard. Their feedback helped me consider the needs of the people who would use such a tool and improve its clarity and organisation.

I would like to acknowledge the developers of Apple Numbers and Microsoft Excel, whose spreadsheet functions, charting tools and documentation made it possible to build and test the dashboard, including the statistical functions used for averages, medians, quartiles and conditional counts.

Finally, I express my deepest gratitude to my family for their patience, encouragement, and continuous support throughout my academic journey.

Developing the Student Performance Dashboard has been a valuable learning experience. It has strengthened my understanding of data modelling, spreadsheet formulas, descriptive statistics, outlier detection, dashboard design and data-driven decision making. I hope the experience gained will serve as a strong foundation for my future work in data analytics.

Aditya Aman
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ABSTRACT

The Student Performance Dashboard is a data analytics workbook, built in Apple Numbers, that converts raw student marks and attendance records into a structured, self-updating view of academic performance. In many institutions, results are maintained in separate registers or plain spreadsheets, which makes it difficult to see class-level trends, identify weak subjects, find students who need support, and compare groups quickly. The dashboard addresses these challenges by placing all records in one data sheet and deriving every summary from that sheet through formulas.

The workbook stores, for each student, a roll number, name, class, marks in four subjects (Mathematics, Science, English and Computer), and attendance percentage. Calculated columns produce total marks, percentage, result, average marks, and a performance band (Excellent, Very Good, Good, Needs Improvement or At Risk). A student passes only when every subject mark is at least 33, and the bands follow fixed percentage thresholds. The workbook has the capacity for up to 200 student records.

Supporting sheets provide a live analysis summary, a PivotTable-style class-by-subject summary, an outlier analysis using the interquartile-range (IQR) method, and an insights sheet that suggests follow-up actions. The Dashboard sheet presents key performance indicators (total students, class average, pass rate, average attendance and at-risk students), an average-marks-by-subject chart, a performance-band chart, a percentage trend for the first 20 records, and quick insights.

For the sample dataset of 31 students used in this project, the class average is 66.0%, the median is 63.8%, the pass rate is 83.9%, and the average attendance is 84.7%. Science is the strongest subject (average 68.7) and Computer the weakest (average 62.0). Five students have at least one subject below the pass mark, eight students have attendance below 75%, and no student is in the At Risk band or is flagged as an IQR outlier. The results were cross-checked by recomputing the figures independently from the extracted data.

The project demonstrates the practical use of data preparation, descriptive statistics, formula-driven modelling, outlier detection and dashboard design. It also provides a foundation for future enhancements such as term-wise trend analysis, automated data validation, a larger real dataset, and integration with a database or a business-intelligence tool.

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1. INTRODUCTION

Educational institutions generate a large amount of performance data every term. Marks in different subjects, attendance records and class details are collected for every student, but the data is often stored in separate registers or in plain tables that are difficult to read at a glance. Teachers and coordinators who want to know how a class is performing, which subject needs attention, or which students need support usually have to calculate these answers manually.

The Student Performance Dashboard was developed to solve this problem with a single, formula-driven workbook. All student records are entered in one data sheet. Every total, percentage, result, performance band, summary table, chart and recommendation is calculated automatically from that sheet, so that adding or correcting a record updates the entire dashboard.

From a technical perspective, the project is a data analytics workbook rather than a programmed application. It is built in Apple Numbers using spreadsheet formulas such as SUM, AVERAGE, MEDIAN, MODE.SNGL, MIN, MAX, IF, COUNTIF and QUARTILE. Charts, KPI tiles and formatted tables present the results visually. The workbook is organised into separate sheets for data, analysis and presentation, which keeps the design easy to understand and maintain.

1.1. Problem Statement

Student results are commonly kept in raw tables without any consistent analysis. It is difficult to identify the weakest subject, compare classes, find students who failed a subject, notice poor attendance, or detect unusual marks that may be data-entry errors. Manual calculation is time-consuming and error-prone, and the results quickly become outdated when marks are corrected or new students are added. Therefore, a structured, self-updating tool is needed that applies consistent pass rules and performance bands, summarises performance by subject and class, and presents the main findings on a single dashboard.

1.2. Objectives

- To store student marks, attendance and class details in one organised data sheet with capacity for up to 200 records.
- To calculate total marks, percentage, average marks, result and performance band automatically for every student.
- To apply a clear pass rule (every subject mark at least 33) and fixed performance bands (Excellent ≥ 90, Very Good ≥ 75, Good ≥ 60, Needs Improvement ≥ 45, At Risk < 45).
- To summarise performance by subject and by class using a PivotTable-style summary.
- To identify unusual marks using the interquartile-range (IQR) outlier method.
- To present key performance indicators, charts and quick insights on a single Dashboard sheet.
- To generate insights and suggested actions, such as the weakest subject and students with low attendance.

1.3. Scope of the Project

The project covers a spreadsheet-based dashboard for monitoring academic performance of a class or institution. Its scope includes data entry for up to 200 students, four subjects, attendance tracking, automatic calculation of results and bands, subject and class summaries, outlier analysis, insights, and a visual dashboard. The workbook uses a sample dataset of 31 records for demonstration. Features such as multi-term trend analysis, automatic import from a college information system, role-based access, and a web-based interface are outside the current scope and are listed as future enhancements.

1.4. Existing System and Proposed System

Aspect	Existing Manual / Basic Methods	Proposed Student Performance Dashboard
Calculation	Totals and percentages are calculated by hand or with separate formulas	Totals, percentages, results and bands are calculated automatically
Consistency	Pass rules and grading may be applied differently by different people	One pass rule and one set of performance bands are applied to all records
Summaries	Class and subject summaries must be prepared separately	Live analysis and PivotTable-style summaries update automatically
Visibility	Results are hidden inside long tables	Dashboard shows KPIs, charts and quick insights on one page
Data quality	Unusual marks may go unnoticed	IQR outlier analysis flags marks for review
Follow-up	Weak subjects and low attendance are found only by manual checking	Insights sheet lists findings and suggested actions

1.5. Organisation of the Report

Chapter 1 introduces the project and its objectives. Chapter 2 describes the hardware, software, functional and non-functional requirements. Chapter 3 explains the system design, tools and modules. Chapter 4 presents the data design and calculation rules. Chapter 5 explains the architecture and flow chart. Chapter 6 discusses data quality, testing and deployment. Chapter 7 presents the output, charts and observations. Chapter 8 concludes the report and discusses future enhancements. Chapter 9 lists the references, and Annexure A gives the complete sample dataset.

2. REQUIREMENTS

2.1. Hardware Requirements

- Processor: Intel Core i3 / Apple silicon / AMD equivalent or higher.
- RAM: Minimum 4 GB; 8 GB recommended for smooth chart rendering.
- Storage: Less than 5 MB for the workbook; additional space for exports and backups.
- Display: Standard laptop or desktop display.
- Network: Not required for using the workbook; internet is useful for updates and sharing.

2.2. Software Requirements

Software	Purpose
macOS, iPadOS or iOS with Apple Numbers	Primary application used to build and edit the workbook (.numbers file)
Microsoft Excel (optional)	Alternative for opening the workbook; best compatibility is noted in the workbook instructions
Spreadsheet functions (SUM, AVERAGE, MEDIAN, MODE.SNGL, MIN, MAX, IF, COUNTIF, QUARTILE, IFERROR)	Calculation of totals, statistics, results, bands and outliers
Numbers charts (bar, doughnut, line)	Visual presentation of subject averages, performance bands and percentage trend
PDF viewer / word processor	Preparation and viewing of this report

2.3. Functional Requirements

- The user can enter or edit student records (roll number, name, class, four subject marks and attendance).
- The system calculates total marks, percentage, average marks, result and performance band for each student.
- The system applies the pass rule: a student passes only when all subject marks are at least 33.
- The system summarises average marks by subject, class-wise averages and student counts.
- The system calculates the mean, median, mode, highest and lowest percentage.
- The system counts students in each performance band and students with attendance below 75%.
- The system identifies outliers in each subject using the IQR method.
- The system displays KPIs, charts, quick insights and suggested actions on the Dashboard and Insights sheets.

2.4. Non-Functional Requirements

- Usability: the dashboard should be clear, colour-coded and easy to read.
- Accuracy: formulas must calculate results consistently for all records.
- Maintainability: data, calculations and presentation should be separated into different sheets.
- Scalability: the workbook should accept up to 200 student records without redesign.
- Portability: the workbook should be shareable and open in common spreadsheet applications.
- Privacy: real student data should be anonymised or protected; the sample dataset contains no real student information.

3. MODEL / SYSTEM DESIGN

3.1. System Overview

The Student Performance Dashboard follows an input–process–output model. The Student Data sheet is the single source of input. Calculated columns and analysis sheets process the data using formulas, and the Dashboard sheet presents the results. Because every output is linked to the data sheet, no figure is typed manually and the dashboard remains consistent when records change.

3.2. Technologies Used

Component	Technology	Role
Workbook	Apple Numbers	Data storage, formulas, charts and layout
Calculations	Spreadsheet formulas	Totals, percentages, results, bands, statistics
Statistics	MEDIAN, MODE.SNGL, QUARTILE	Central tendency and IQR outlier bounds
Visualisation	Bar, doughnut and line charts	Subject averages, band distribution, percentage trend
Summaries	PivotTable-style formula tables	Class-by-subject averages and student counts
Reporting	Word processor / PDF	Project report preparation

3.3. Main Modules

The workbook is organised into the following sheets, each acting as a module:

- **Dashboard:** KPI tiles (total students, class average, pass rate, average attendance, at-risk students), charts, a performance filter and quick insights.
- **Student Data:** the input sheet. Columns A–H are entered by the user; columns I–M are calculated and prefilled through row 201.
- **Live Analysis Summary:** average marks by subject, mean, median, mode, highest and lowest percentage, and the number of students in each performance band.

- **Pivot Summary:** a PivotTable-style summary of subject averages, average percentage and student count for each class, plus overall subject averages and the number of students below 33.
- **Outlier Analysis:** IQR lower and upper bounds for each subject, outlier counts and student-level flags.
- **Insights:** live findings with suggested interpretation or action.
- **How to Use:** step-by-step instructions for editing the data and reading the dashboard.
- **Supporting sheets:** Filter Data and Performance Details, which support filtered and per-student views.

3.4. Workflow

1. The user opens the workbook and goes to the Student Data sheet.
2. The user enters or edits columns A–H for a student (roll number, name, class, four marks and attendance).
3. The calculated columns I–M compute total, percentage, result, average and performance band.
4. The analysis sheets read the calculated data and recompute summaries, class-wise averages and outlier bounds.
5. The Insights sheet updates the findings and suggested actions.
6. The Dashboard displays the updated KPIs, charts and quick insights.

3.5. Design Considerations

The workbook separates data entry, calculation, analysis and presentation, which improves maintainability. Formula columns are prefilled so that new students can be added by typing only into the input columns. The pass rule and band thresholds are kept in formulas that can be edited if an institution uses a different rule. A PivotTable-style summary built with formulas is used instead of a native PivotTable for compatibility with Apple Numbers. Outliers are treated as items to review, not as automatic errors.

4. DATA DESIGN

The project uses a tabular data model. The Student Data sheet is the only input table; all other sheets are derived from it. Each row represents one student, and each column represents one attribute or calculated value.

4.1. Student Data Sheet

Col.	Field	Type	Description / Rule
A	Roll No	Text	Unique student identifier (e.g., STU001)
B	Student Name	Text	Name of the student (sample names)
C	Class	Text	Class 10, Class 11, Class 12 (one record is classified as MCA)
D	Mathematics	Number	Marks out of 100

E	Science	Number	Marks out of 100
F	English	Number	Marks out of 100
G	Computer	Number	Marks out of 100
H	Attendance %	Number	Attendance percentage (0–100)
I	Total Marks	Calculated	Sum of the four subject marks (out of 400)
J	Percentage %	Calculated	Total marks ÷ 4, shown to one decimal place
K	Result	Calculated	Pass if every subject mark ≥ 33; otherwise Fail
L	Average Marks	Calculated	Average of the four subject marks
M	Performance Band	Calculated	Band assigned from the percentage

4.2. Calculation Rules

Rule	Definition
Pass rule	A student passes only when all four subject marks are at least 33
Excellent	Percentage ≥ 90
Very Good	Percentage ≥ 75 and < 90
Good	Percentage ≥ 60 and < 75
Needs Improvement	Percentage ≥ 45 and < 60
At Risk	Percentage < 45
Outlier (IQR)	A mark below Q1 − 1.5 × IQR or above Q3 + 1.5 × IQR, where IQR = Q3 − Q1
Low attendance	Attendance below 75%

4.3. Sample Dataset Overview

The sample dataset contains 31 records: 12 students in Class 10, 10 in Class 11, 8 in Class 12, and one record classified as MCA. Subject marks range from 29 to 100, and attendance ranges from 66% to 99%. The complete dataset is given in Annexure A. The data is for demonstration only and does not represent real students.

4.4. Analysis Sheet Structures

- **Live Analysis Summary:** two tables — subject averages, and metrics (mean, median, mode, highest and lowest percentage) — plus a table of student counts for each performance band.
- **Pivot Summary:** class rows (Class 10, 11, 12) against subject columns, with average percentage and student count, and an overall table of subject averages and students below 33.
- **Outlier Analysis:** lower bound, upper bound and outlier count per subject, followed by student-level flags and an interpretation column.
- **Insights:** finding, live result, and suggested interpretation or action.

5. SYSTEM ARCHITECTURE DESIGN AND FLOW CHART

5.1. Architecture Overview

The workbook uses a three-layer architecture consisting of an input layer, a calculation and analysis layer, and a presentation layer. Data flows in one direction: from the Student Data sheet, through the calculated columns and analysis sheets, to the Dashboard. This one-way flow prevents circular dependencies and keeps every figure traceable to its source.

5.2. Input Layer

The Student Data sheet accepts the student's roll number, name, class, four subject marks and attendance. Only columns A–H are edited by the user. The workbook instructions advise keeping the headers unchanged and keeping marks and attendance between 0 and 100.

5.3. Calculation and Analysis Layer

Columns I–M compute totals, percentages, results, averages and bands. The Live Analysis Summary, Pivot Summary, Outlier Analysis and Insights sheets then read these columns and compute subject averages, class-wise averages, central-tendency measures, band counts, outlier bounds and recommended actions.

5.4. Presentation Layer

The Dashboard sheet presents five KPI tiles, an average-marks-by-subject chart, a performance-band chart, a line chart of percentages for the first 20 records, a performance filter and a quick-insights panel. Colour coding is used to separate the KPIs.

5.5. Flow Chart

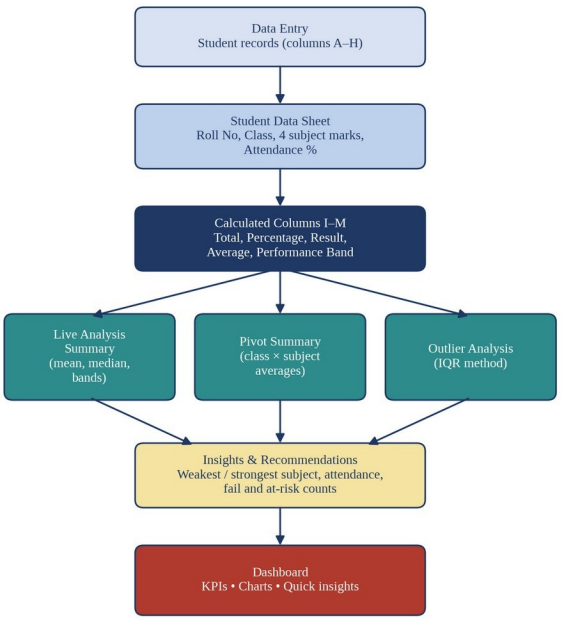


Figure 5.1: Data flow of the Student Performance Dashboard

6. DATA QUALITY, TESTING AND DEPLOYMENT

6.1. Data Quality and Integrity Measures

- **Single source of truth:** all summaries are derived from the Student Data sheet, so there is no duplicated or manually typed output.
- **Protected formulas:** columns I–M are prefilled; users type only into columns A–H.
- **Consistent rules:** one pass rule and one set of band thresholds are used for every record.
- **Range guidance:** marks and attendance are expected to be between 0 and 100.
- **Outlier review:** the IQR method highlights unusual marks for review, helping to find data-entry mistakes.
- **Sample data:** the dataset is fictional, which avoids exposing real student information.

6.2. Testing

Testing checked whether the calculated columns, summaries and KPIs behave as expected. The values stored in the workbook were extracted and the main statistics were recomputed independently, then compared with the workbook's results. The tests below should be repeated after any change to the data or formulas.

6.3. Test Cases

No.	Test Case	Expected Result	Observed Result
1	Total marks for STU001 (66, 74, 65, 50)	Total = 255	255 — matches
2	Percentage for STU001	255 ÷ 4 = 63.75, shown as 63.8	63.8 — matches
3	Pass rule for STU002 (Mathematics 31)	Result = Fail	Fail — matches
4	Band for STU020 (89.0%) and STU004 (79.8%)	Both Very Good (75 ≤ % < 90)	Very Good — matches
5	Add a new student in the next blank row	Total students and averages update	31 students counted; KPIs updated
6	Class average, pass rate and average attendance	65.98%, 83.87% (26 of 31), 84.68%	Dashboard shows 66.0%, 83.9%, 84.7%
7	Subject averages	Math 65.87, Science 68.74, English 67.23, Computer 61.97	Live Analysis matches
8	Students below 33 in any subject	5 students (2 Math, 2 English, 1 Computer)	Insights shows 5
9	Students with attendance below 75%	8 students	Insights shows 8
10	IQR outlier check	No mark outside the IQR bounds	0 outliers reported
11	Class-wise pivot student counts	12 + 10 + 8 = 30 (Class 10/11/12)	30 — the MCA record is not included in the class pivot
12	Mode % metric	A valid mode of the percentages	Shows 0 (see Section 6.4)

6.4. Limitations Identified During Testing

- The Mode % cell in the Live Analysis Summary shows 0 because the MODE.SNGL formula refers to the full range of 200 rows, including blank formula rows, and falls back to 0 through IFERROR. Computed directly from the 31 percentages, the values 74.8% and 79.8% each occur twice, so the data is bimodal.
- The Pivot Summary lists only Class 10, Class 11 and Class 12, so the single MCA record is counted in the dashboard totals but is not shown in the class-wise summary.
- The dataset is small and fictional; conclusions are illustrative and should not be applied to real students.
- The performance filter and charts depend on the application; Apple Numbers and Excel may render some charts differently.

6.5. Deployment and Usage

The dashboard is deployed as a single workbook file. To use it, open the file in Apple Numbers (or Microsoft Excel), go to the Student Data sheet, and replace or add student records in columns A–H. Formula columns I–M and all analysis sheets update automatically, and the results can be viewed on the Dashboard. The file can be shared through cloud storage or exported to PDF or Excel format for circulation.

7. OUTPUT

This chapter presents the output of the workbook. Figure 7.1 is the Dashboard sheet exported from the workbook, and the remaining figures present the analysis results in chart form using the data in the workbook.

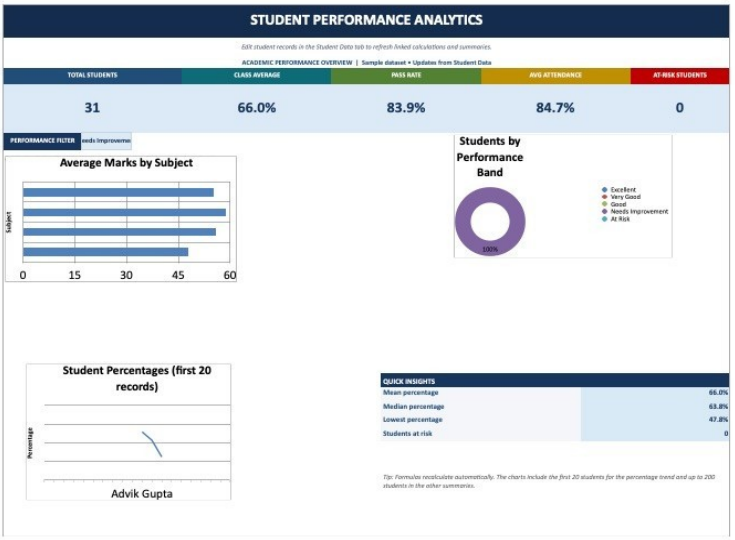


Figure 7.1: Dashboard sheet (export preview from the workbook)

7.1. Key Performance Indicators

KPI	Value
Total students	31
Class average (mean percentage)	66.0%
Median percentage	63.8%
Highest / lowest percentage	89.0% (STU020) / 47.8% (STU028)
Pass rate	83.9% (26 of 31 students)
Average attendance	84.7%
Students below 75% attendance	8
Students at risk (percentage below 45)	0

7.2. Subject-wise Performance

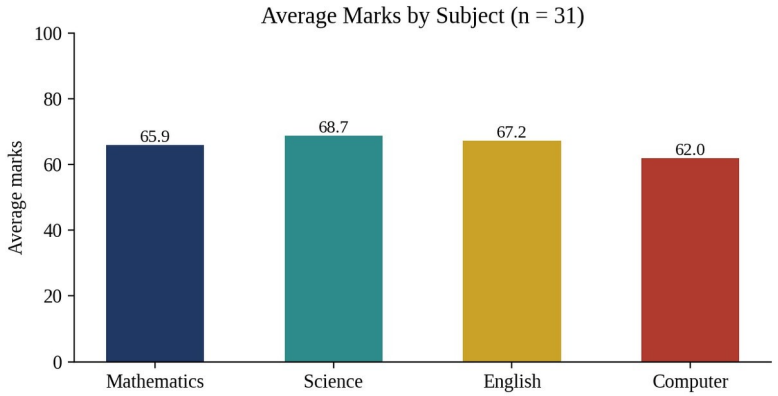


Figure 7.2: Average marks by subject

Science has the highest average (68.7), followed by English (67.2) and Mathematics (65.9). Computer has the lowest average (62.0) and is identified by the Insights sheet as the weakest subject.

7.3. Performance Band Distribution

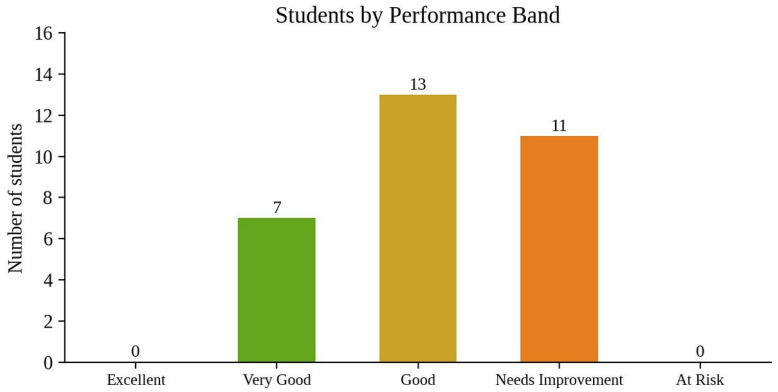


Figure 7.3: Students by performance band

Of the 31 students, 7 are Very Good, 13 are Good and 11 need improvement. No student is Excellent or At Risk.

7.4. Class-wise Performance

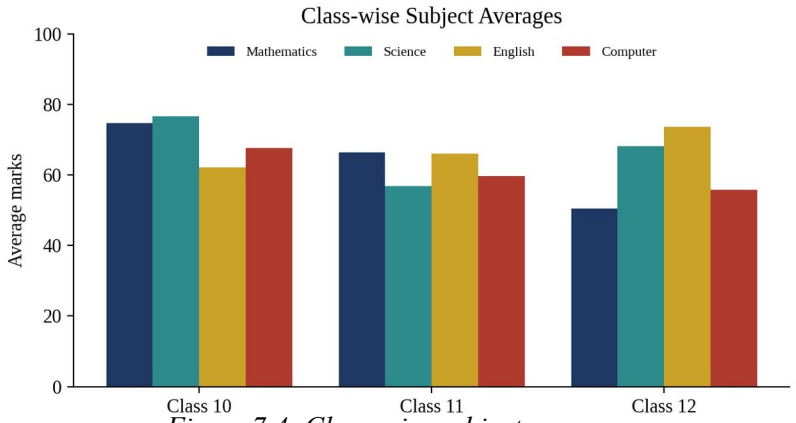


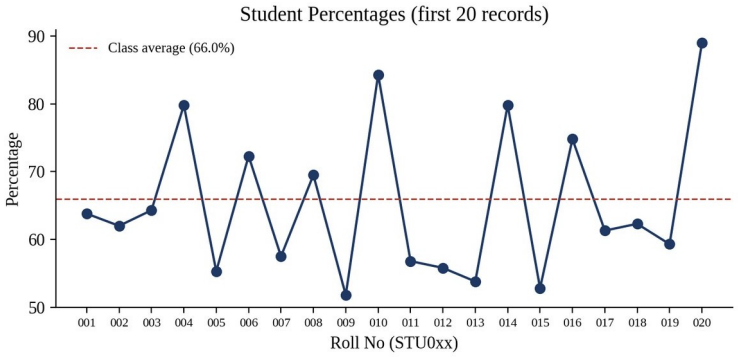
Figure 7.4: Class-wise subject averages

Class	Mathematics	Science	English	Computer	Average %	Students
Class 10	74.75	76.58	62.08	67.67	70.29	12
Class 11	66.30	56.90	66.10	59.60	62.26	10
Class 12	50.50	68.13	73.63	55.75	62.05	8

Class 10 has the highest average percentage (70.3%). Class 12 has the weakest Mathematics average (50.5), and Class 11 has the weakest Science average (56.9). English is the weakest subject for Class 10 (62.1).

7.5. Percentage Trend

Figure 7.5: Student percentages for the first 20 records



7.6. Attendance and Performance

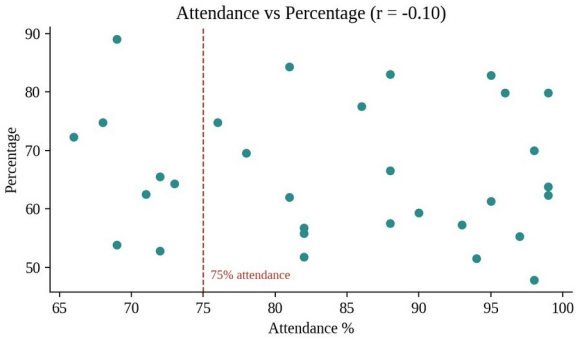


Figure 7.6: Attendance versus percentage

The correlation between attendance and percentage in this sample is −0.10, which indicates practically no linear relationship. With only 31 records, this should not be generalised.

7.7. Students Needing Attention

Table 7.1: Students with a subject below the pass mark (33)

Roll No	Name	Class	Mathematics	Science	English	Computer
STU002	Diya Gupta	Class 10	31 *	59	77	81
STU012	Kiara Patel	Class 10	34	72	30 *	87
STU025	Dhruv Singh	Class 12	29 *	52	98	50
STU027	Veer Mehta	Class 10	65	61	51	29 *
STU030	Aanya Sharma	Class 11	96	56	31 *	79

** Subject below 33.*

Table 7.2: Students with attendance below 75%

Roll No	Name	Class	Attendance %	Percentage %
STU006	Ishita Mehta	Class 12	66	72.3
STU016	Myra Verma	Class 10	68	74.8
STU013	Rohan Mehta	Class 12	69	53.8
STU020	Navya Mehta	Class 10	69	89
STU029	Om Sharma	Class 10	71	62.5
STU015	Krish Joshi	Class 11	72	52.8
STU030	Aanya Sharma	Class 11	72	65.5
STU003	Kabir Khan	Class 11	73	64.3

Table 7.3: Top five students by percentage

Rank	Roll No	Name	Class	Percentage %	Band
1	STU020	Navya Mehta	Class 10	89	Very Good
2	STU010	Saanvi Verma	Class 12	84.3	Very Good
3	STU031	Aditya Aman	MCA	83	Very Good
4	STU026	Riya Mehta	Class 11	82.8	Very Good
5	STU004	Anaya Verma	Class 10	79.8	Very Good

7.8. Observations

- The class is mostly in the Good and Needs Improvement bands, and the class average of 66.0% is higher than the median of 63.8%, which suggests that a few high scorers lift the average.
- Five students (STU002, STU012, STU025, STU027 and STU030) fail because of a single subject mark below 33; targeted support in that subject could change their result.
- Eight students have attendance below 75%, so attendance follow-up is recommended.
- No IQR outliers were found, which suggests that the marks do not contain extreme or obviously mistyped values.
- The Insights sheet recommends reviewing Computer, the weakest subject, through extra practice or revision sessions.

8. CONCLUSION

8.1. Conclusion

The Student Performance Dashboard demonstrates how a well-structured spreadsheet can serve as a compact data analytics solution. By keeping all inputs in one sheet and deriving every result from formulas, the workbook gives consistent totals, percentages, results and performance bands, and summarises performance by subject and class without manual calculation.

The sample analysis shows how the dashboard supports decisions. It identifies Computer as the weakest subject, Science as the strongest, five students who need subject-level support, and eight students with low attendance. The IQR analysis confirms that no extreme marks are present in the dataset, and independent recomputation confirmed the main results.

The project provided practical experience in data preparation, descriptive statistics, formula design, outlier detection, dashboard layout and interpretation of results. The limitations identified in testing, such as the Mode % cell and the class summary not covering the MCA record, show where the workbook can be improved. The findings are illustrative because the dataset is small and fictional.

8.2. Future Enhancements

- Track multiple terms or exams to show performance trends over time.
- Add data validation so that marks and attendance outside 0–100 are rejected at entry.
- Make the Pivot Summary dynamic so that any class name, including MCA, appears automatically.
- Correct the Mode % formula so that it ignores blank rows.
- Add more subjects, student-wise report cards and comparison between sections.
- Use a larger real (anonymised) dataset and test correlation between attendance and marks.
- Connect the data to a database or a business-intelligence tool such as Power BI or Tableau, and add predictive models for identifying at-risk students.

9. REFERENCES

1. Apple Inc. Numbers User Guide. https://support.apple.com/guide/numbers/
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ANNEXURE A: STUDENT DATASET

The table below lists the sample records stored in the Student Data sheet, together with the calculated columns.

Roll No	Name	Class	Math	Sci	Eng	Comp	Att %	Total	%	Result	Band
STU001	Aarav Khan	Class 11	66	74	65	50	99	255	63.8	Pass	Good
STU002	Diya Gupta	Class 10	31	59	77	81	81	248	62	Fail	Good
STU003	Kabir Khan	Class 11	79	45	98	35	73	257	64.3	Pass	Good
STU004	Anaya Verma	Class 10	96	99	54	70	99	319	79.8	Pass	Very Good
STU005	Vivaan Singh	Class 12	36	67	80	38	97	221	55.3	Pass	Needs Improvement
STU006	Ishita Mehta	Class 12	46	80	92	71	66	289	72.3	Pass	Good
STU007	Arjun Das	Class 11	34	73	34	89	88	230	57.5	Pass	Needs Improvement
STU008	Meera Joshi	Class 10	79	58	42	99	78	278	69.5	Pass	Good
STU009	Reyansh Verma	Class 12	74	35	62	36	82	207	51.8	Pass	Needs Improvement
STU010	Saanvi Verma	Class 12	99	96	98	44	81	337	84.3	Pass	Very Good
STU011	Advik Gupta	Class 11	68	49	70	40	82	227	56.8	Pass	Needs Improvement
STU012	Kiara Patel	Class 10	34	72	30	87	82	223	55.8	Fail	Needs Improvement
STU013	Rohan Mehta	Class 12	34	53	67	61	69	215	53.8	Pass	Needs Improvement
STU014	Aditi Mehta	Class 10	100	95	45	79	96	319	79.8	Pass	Very Good
STU015	Krish Joshi	Class 11	51	53	60	47	72	211	52.8	Pass	Needs Improvement
STU016	Myra Verma	Class 10	96	67	68	68	68	299	74.8	Pass	Good
STU017	Ishaan Kaur	Class 11	39	67	60	79	95	245	61.3	Pass	Good
STU018	Sara Singh	Class 12	38	97	58	56	99	249	62.3	Pass	Good
STU019	Dev Gupta	Class 12	48	65	34	90	90	237	59.3	Pass	Needs Improvement
STU020	Navya Mehta	Class 10	98	80	94	84	69	356	89	Pass	Very Good
STU021	Yash Mehta	Class 10	57	89	68	52	88	266	66.5	Pass	Good
STU022	Tara Joshi	Class 10	100	94	82	34	86	310	77.5	Pass	Very Good
STU023	Manav Joshi	Class 10	99	57	68	75	76	299	74.8	Pass	Good
STU024	Pari Verma	Class 11	83	73	85	39	98	280	70	Pass	Good
STU025	Dhruv Singh	Class 12	29	52	98	50	93	229	57.3	Fail	Needs Improvement
STU026	Riya Mehta	Class 11	92	45	96	98	95	331	82.8	Pass	Very Good
STU027	Veer Mehta	Class 10	65	61	51	29	94	206	51.5	Fail	Needs Improvement
STU028	Anvi Patel	Class 11	55	34	62	40	98	191	47.8	Pass	Needs Improvement
STU029	Om Sharma	Class 10	42	88	66	54	71	250	62.5	Pass	Good
STU030	Aanya Sharma	Class 11	96	56	31	79	72	262	65.5	Fail	Good
STU031	Aditya Aman	MCA	78	98	89	67	88	332	83	Pass	Very Good